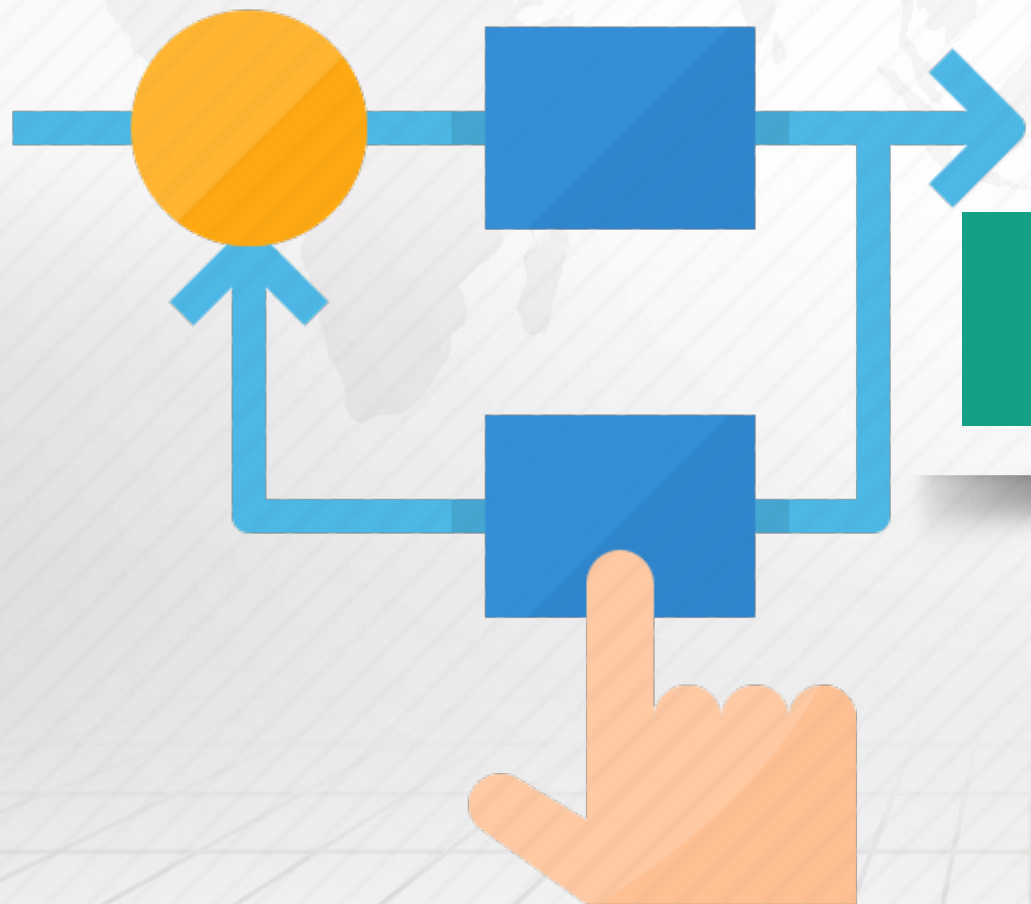




江苏科技大学
JIANGSU UNIVERSITY OF SCIENCE AND TECHNOLOGY



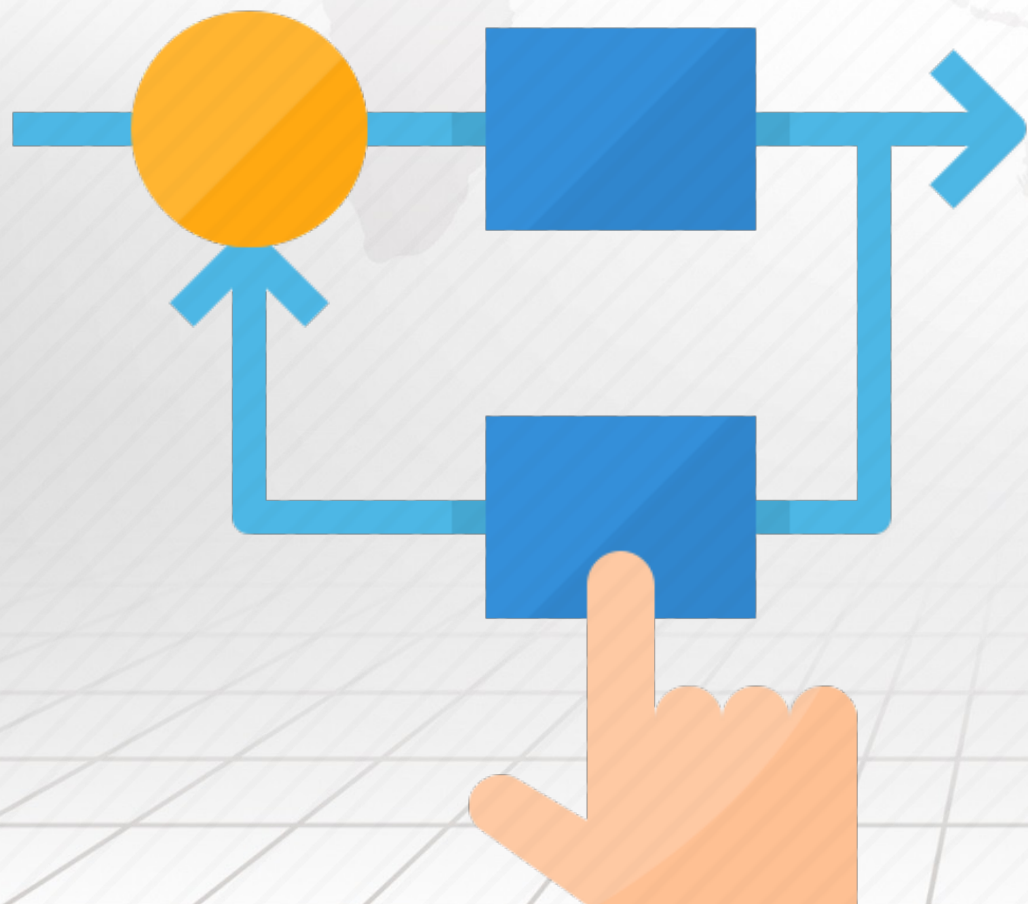
自动控制原理

第一章	第二章	第三章	第四章	第五章	第六章
基本概念	数学模型	时域分析法	根轨迹法	频域分析法	校正方法

19. 幅相特性：换个角度看频域



学习目标



01

能理解开环幅相特性的含义

重

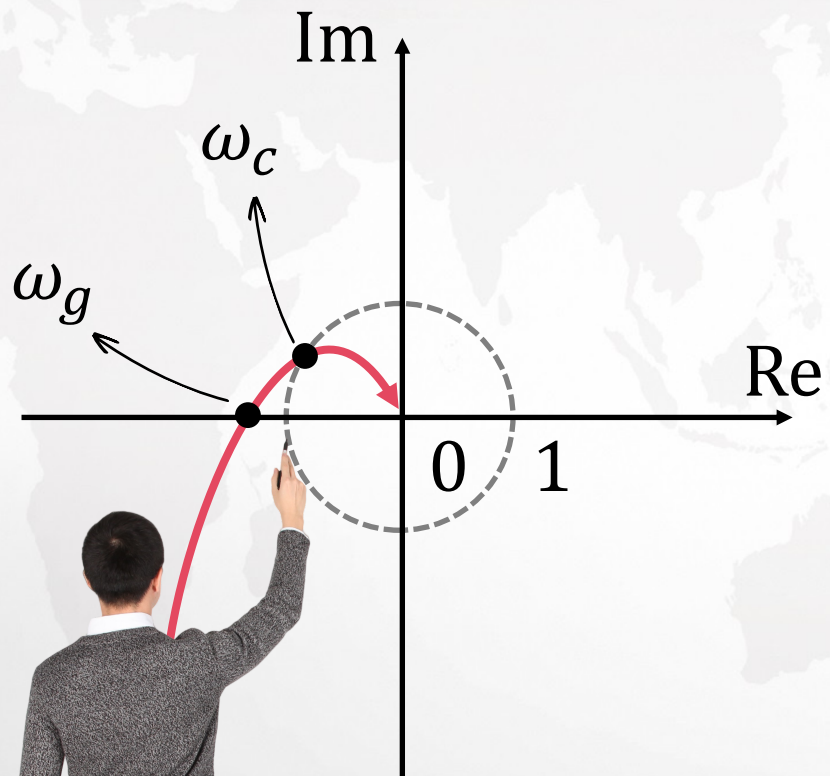
能手工绘制概略开环幅相特性

难

能确定开环幅相特性的特征点

04

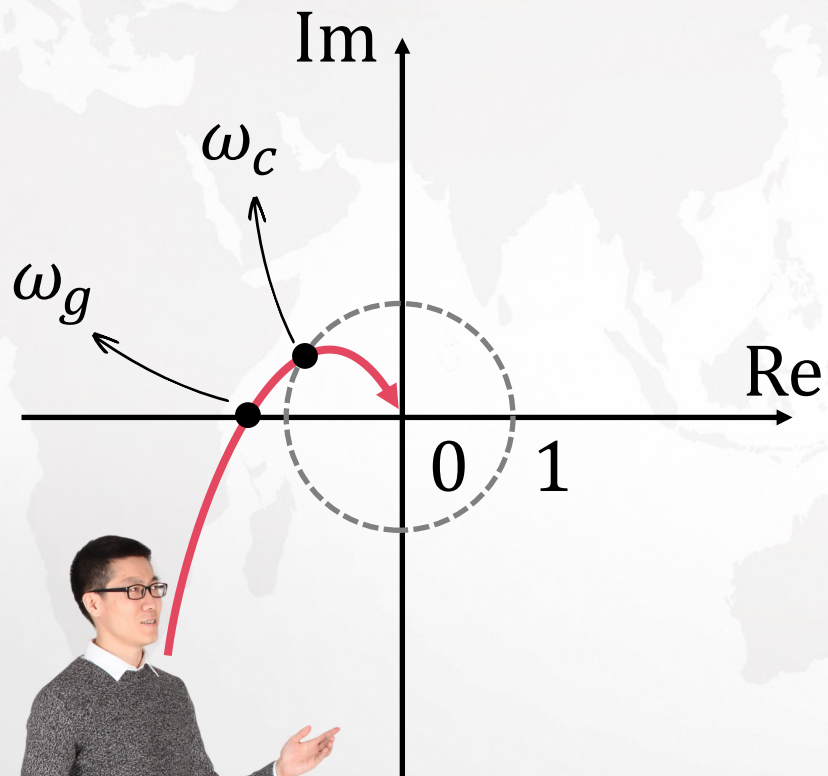
能使用计算机辅助验证



老师，您在
画什么啊？

这叫开环幅相特性，
也叫奈奎斯特图





这个图有什么用啊？

这么简单一条线就能判断？

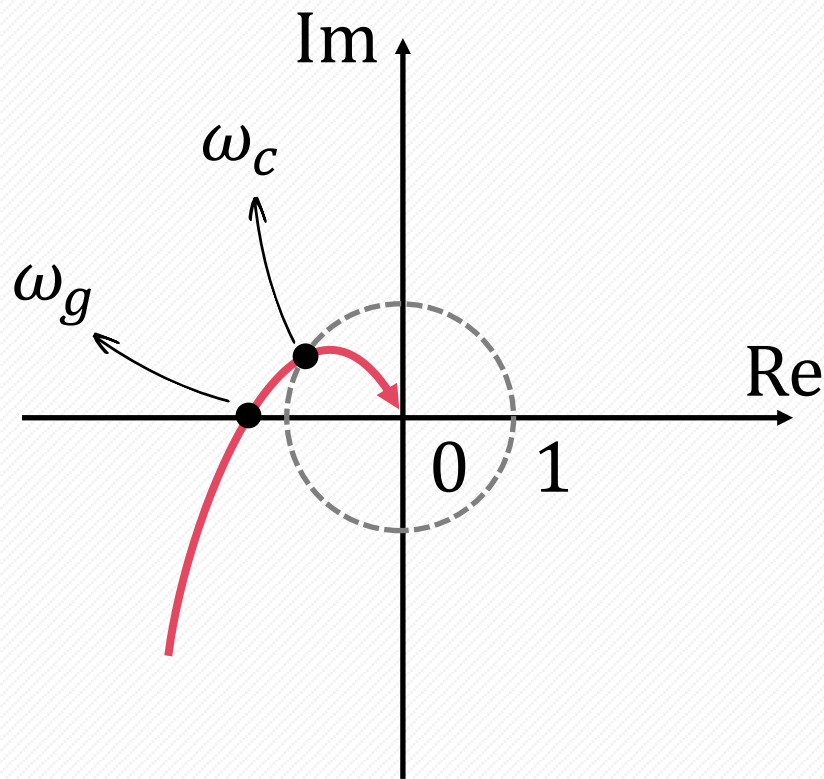
可以用来判断反馈系统的稳定性。



系统开环幅相曲线

定义

当输入信号的频率 $\omega: 0 \rightarrow \infty$ 变化时，频率响应的幅值和相角在极坐标系上形成的轨迹



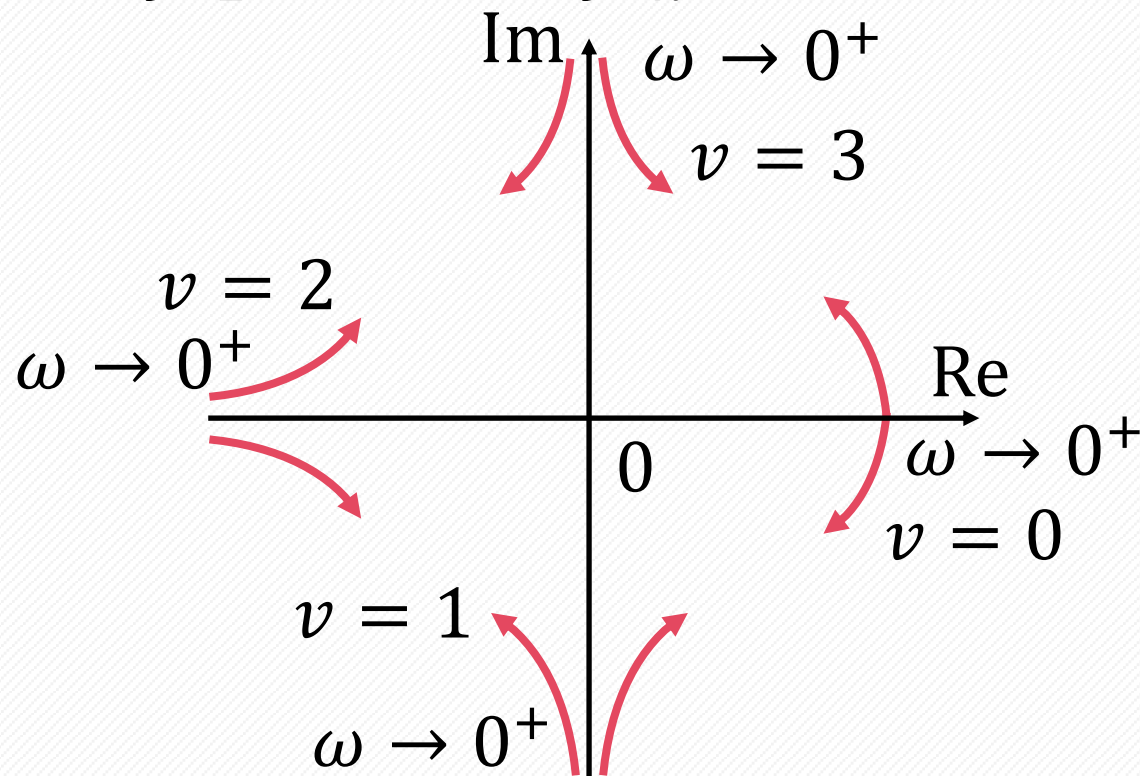
系统开环幅相曲线

$$G(s)H(s) = \frac{K(\tau_1 s + 1)(\tau_2 s + 1) \cdots (\tau_m s + 1)}{s^v (T_1 s + 1)(T_2 s + 1) \cdots (T_n s + 1)} \quad (n > m)$$

$$G(j\omega)H(j\omega) = \frac{K(j\tau_1\omega + 1)(j\tau_2\omega + 1) \cdots (j\tau_m\omega + 1)}{(j\omega)^v (jT_1\omega + 1)(jT_2\omega + 1) \cdots (jT_n\omega + 1)}$$

01 起点(低频段)

$$G(j0^+)H(j0^+) = \lim_{\omega \rightarrow 0^+} \frac{K}{(j\omega)^v}$$



系统开环幅相曲线

02 终点(高频段): $\omega \rightarrow +\infty$

$$G(j\omega)H(j\omega) = \frac{K(j\tau_1\omega + 1)(j\tau_2\omega + 1) \cdots (j\tau_m\omega + 1)}{(j\omega)^v(jT_1\omega + 1)(jT_2\omega + 1) \cdots (jT_n\omega + 1)}$$

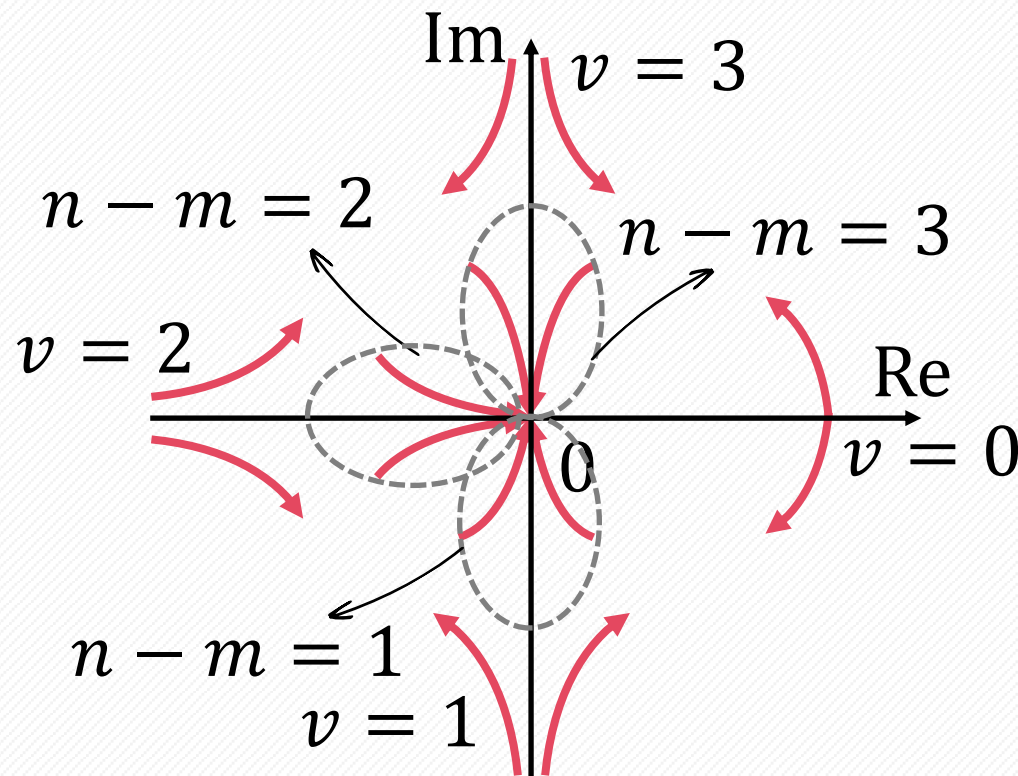
$$\lim_{\omega \rightarrow +\infty} G(j\omega)H(j\omega) = \lim_{\omega \rightarrow +\infty} \frac{K(j\omega)^m}{(j\omega)^n} = \lim_{\omega \rightarrow +\infty} \frac{K}{(j\omega)^{n-m}}, (n > m)$$

终点处幅值

$$\lim_{\omega \rightarrow +\infty} |G(j\omega)H(j\omega)| = 0$$

终点处相角

$$\lim_{\omega \rightarrow +\infty} \angle G(j\omega)H(j\omega) = (n - m)(-\frac{\pi}{2})$$



系统开环幅相曲线

03

与负实轴交点:试探法

方法1

$$\operatorname{Im}[G(j\omega_g)] = 0 \rightarrow \omega_g \rightarrow \operatorname{Re}[G(j\omega_g)]$$

方法2

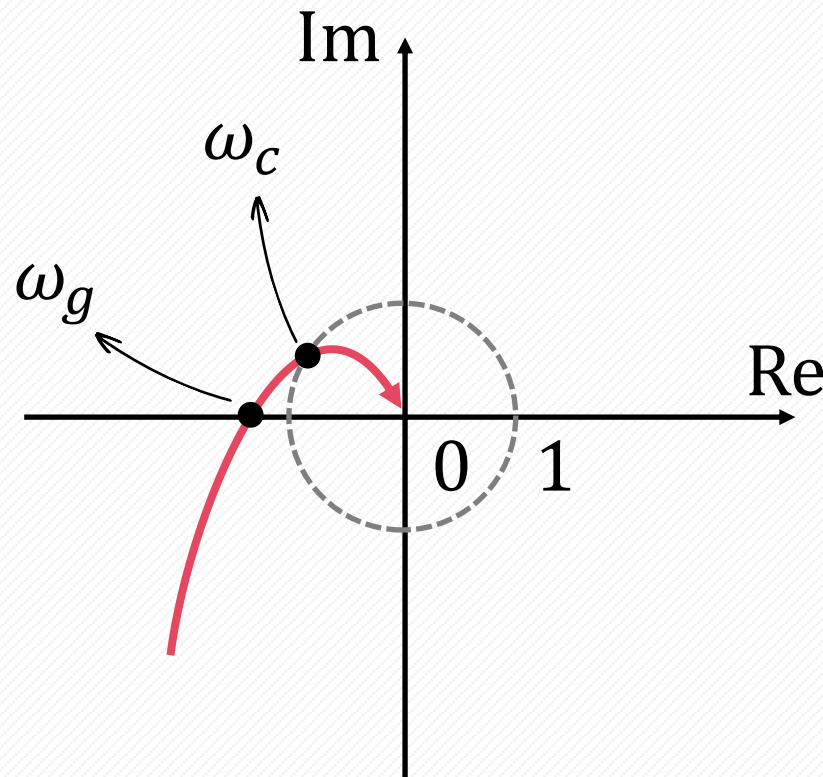
$$\angle G(j\omega_g) = -180^\circ \rightarrow \omega_g \rightarrow |G(j\omega_g)|$$

相角穿越频率

$$\omega_g: \angle G(j\omega_g) = -180^\circ$$

幅值穿越频率 截止频率

$$\omega_c: |G(j\omega_c)| = 1, L(j\omega_c) = 0 \text{ dB}$$



概略绘制极坐标图步骤

01 求频率特性

开环频率特性 $G(j\omega)$

幅频特性 $|G(j\omega)|$

相频特性 $\angle G(j\omega)$

实频特性 $\text{Re}[G(j\omega)]$

虚频特性 $\text{Im}[G(j\omega)]$

02 求特征点

起点

$$|G(j0)| = K, (v = 0), \infty, (v \neq 0)$$
$$\angle G(j0) = v(-90^\circ)$$

终点

$$G(j\infty) = 0 \angle -90^\circ(n - m)$$

负实轴交点

$$\text{Im}[G(j\omega_g)] = 0 \rightarrow \omega_g \rightarrow \text{Re}[G(j\omega_g)]$$

$$\angle G(j\omega_g) = -180^\circ \rightarrow \omega_g \rightarrow |G(j\omega_g)|$$

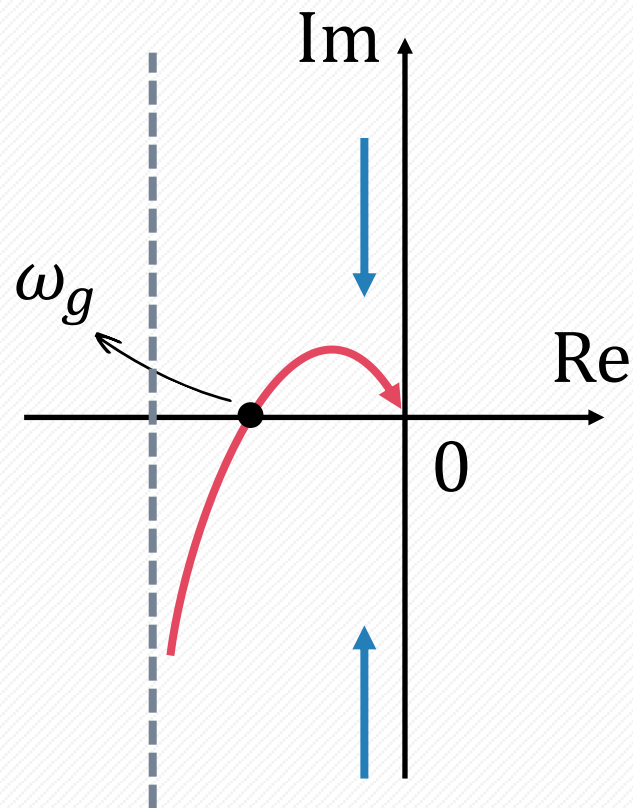
实部渐近线

$$V_x = \lim_{\omega \rightarrow 0} \text{Re}[G(j\omega)], (v = 1)$$

03 概略绘图

不含零点时单调收缩

含零点时可能扭曲



绘制概略极坐标图例1

01 求频率特性

$$G(j\omega) = \frac{10}{(0.1j\omega+1)(j\omega+1)}$$

$$G(s) = \frac{10}{(0.1s+1)(s+1)}$$
$$|G(j\omega)| = \frac{10}{\sqrt{(0.1\omega)^2 + 1}\sqrt{\omega^2 + 1}}$$
$$\angle G(j\omega) = -\arctan 0.1\omega - \arctan \omega$$

02 求特征点

起点

$$G(j0) = 10\angle 0^\circ$$

终点

$$G(j\infty) = 0\angle -180^\circ$$

负实轴交点

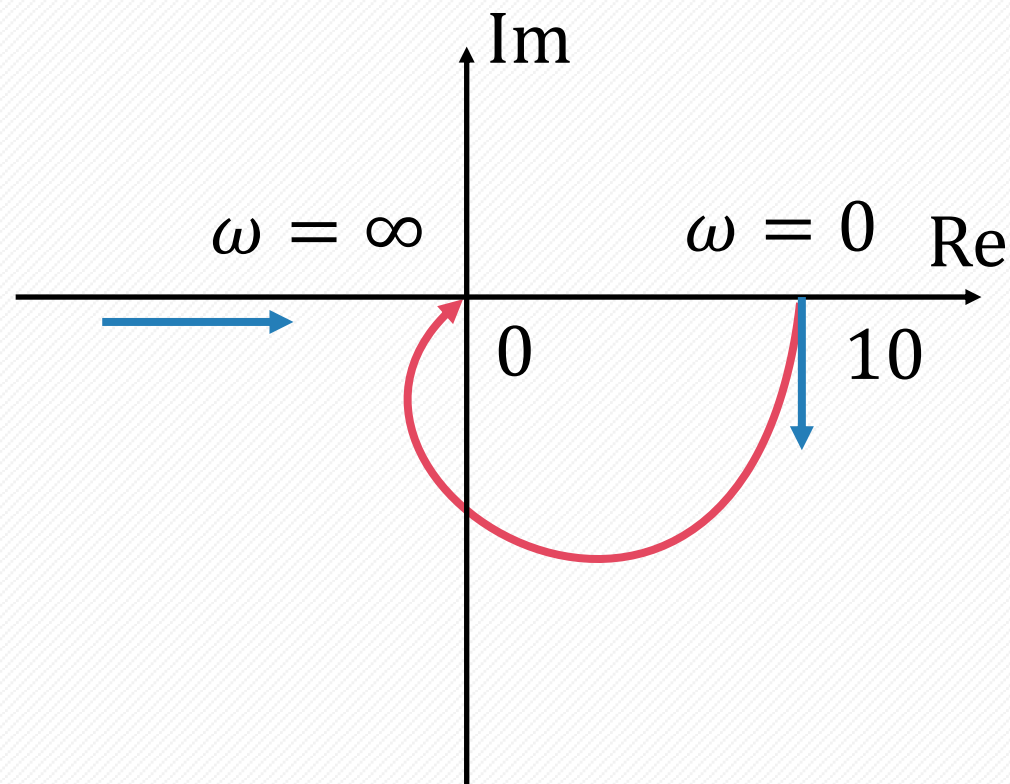
幅角范围 $[0^\circ, -180^\circ]$
不穿越负实轴

实部渐近线

$v = 0$ ，无渐近线

03 概略绘图

不含零点时单调收缩



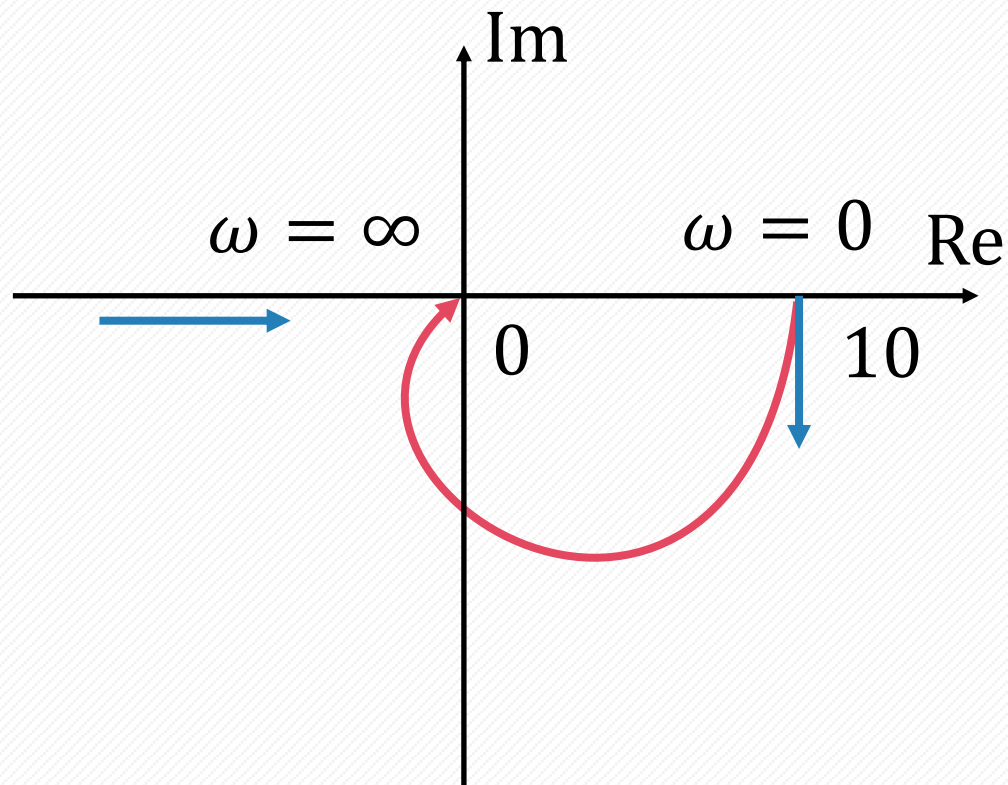
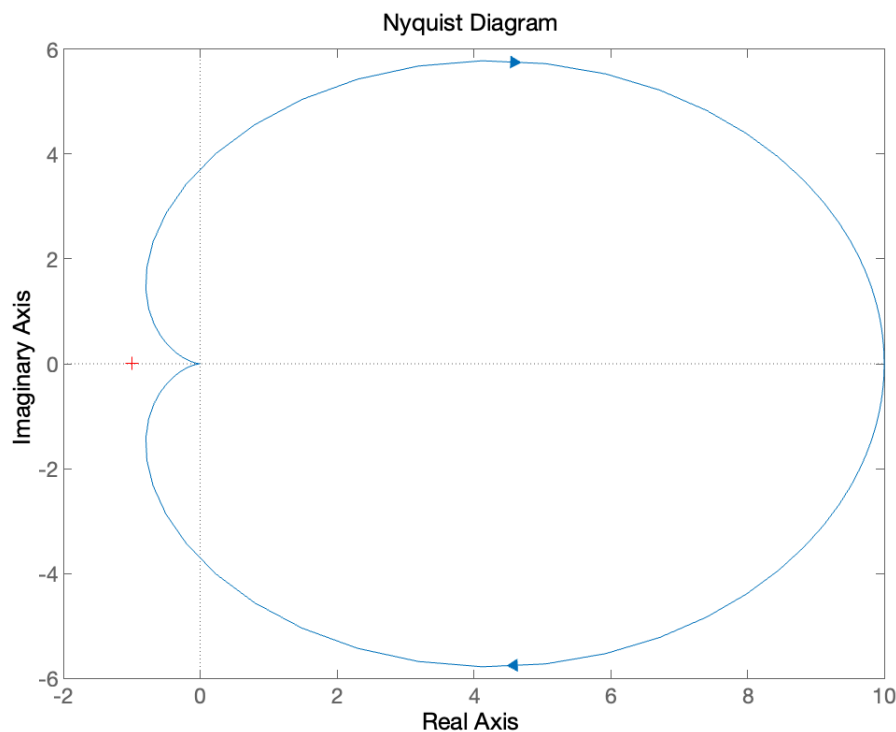
绘制概略极坐标图例1

?

为什么不一样？

代码

```
s = tf('s'); % 建立传递函数的复数变量  
G = 10/((0.2*s+1)*(s+1)); % 建立传递函数  
nyquist(G) % 绘制开环幅相特性
```



绘制概略极坐标图例2

$$G(s) = \frac{10}{s(s+1)}$$

01 求频率特性

$$G(j\omega) = \frac{10}{j\omega(1+j\omega)} = \frac{-10j(1-j\omega)}{-j \times j\omega(1+j\omega)(1-j\omega)} = \frac{-10\omega - 10j}{\omega(\omega^2 + 1)}$$

02 求特征点

起点

$$G(j0^+) = \infty \angle -90^\circ = \frac{-10}{\omega^2 + 1} - j \frac{10}{\omega^3 + \omega}$$

终点

$$G(j\infty) = 0 \angle -180^\circ$$

负实轴交点

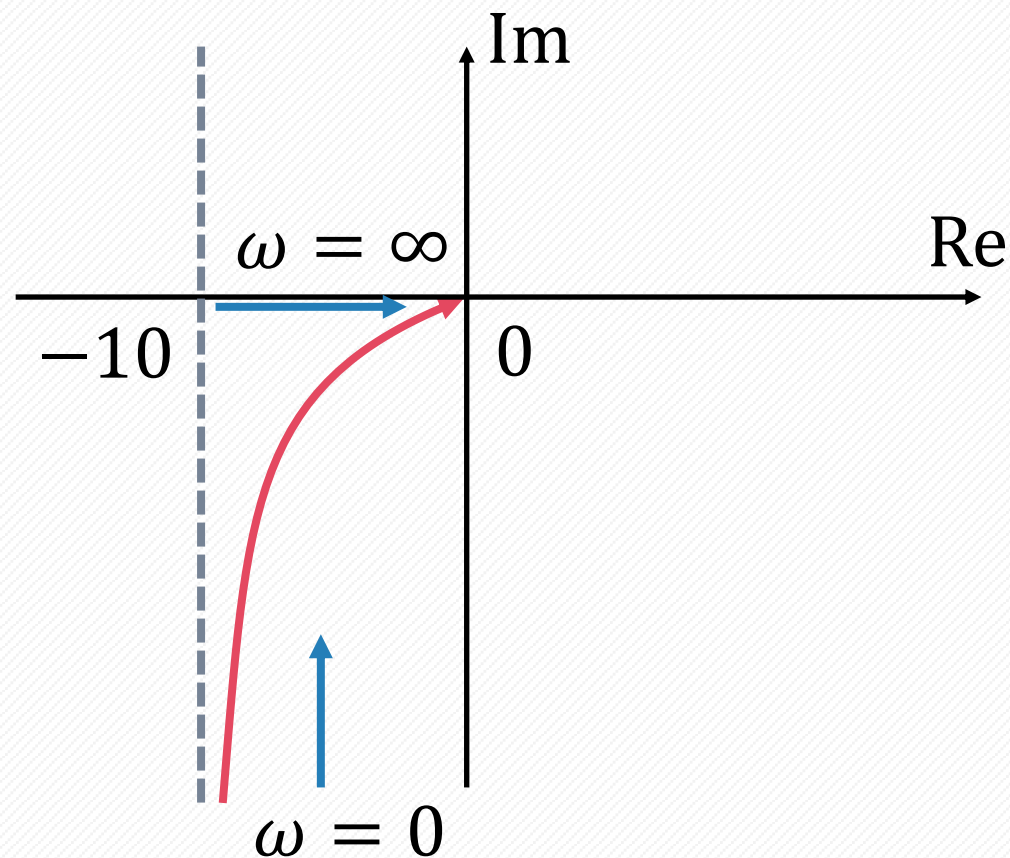
幅角范围 $[-90^\circ, -180^\circ]$
不穿越负实轴

实部渐近线

$$\lim_{\omega \rightarrow 0^+} \frac{-10}{\omega^2 + 1} = -10$$

03 概略绘图

不含零点时单调收缩



绘制概略极坐标图例3

01 求频率特性

$$G(j\omega) = \frac{10}{(j\omega)^2(j\omega+1)}$$



$$|G(j\omega)| = \frac{10}{\omega^2 \sqrt{\omega^2 + 1}}$$

02 求特征点

起点

$$G(j0) = \infty \angle -180^\circ$$

$$\angle G(j\omega) = -180^\circ - \arctan \omega$$

终点

$$G(j\infty) = 0 \angle -270^\circ$$

负实轴交点

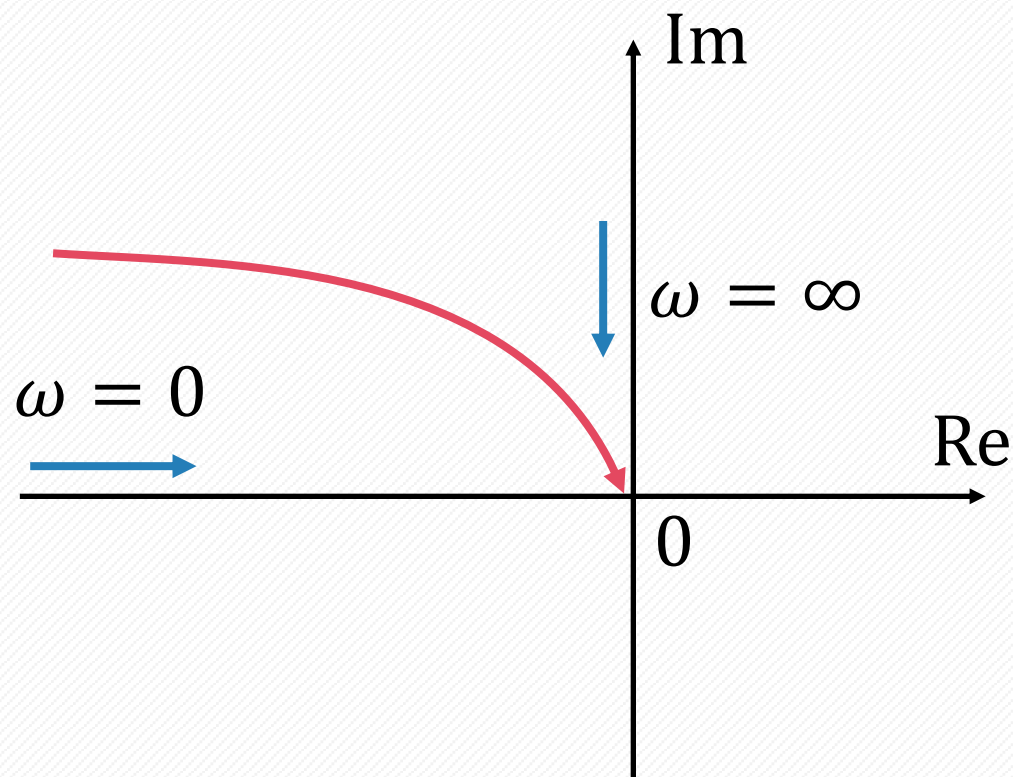
幅角范围 $[-180^\circ, -270^\circ]$
不穿越负实轴

实部渐近线

$v = 2$, 无渐近线

03 概略绘图

不含零点时单调收缩



绘制概略极坐标图例4

01 求频率特性

$$G(j\omega) = \frac{5}{j\omega(1+j\omega)(1+j2\omega)} = \frac{-j5(1-j\omega)(1-j2\omega)}{\omega(1+\omega^2)(1+4\omega^2)}$$

02 求特征点

$$= \frac{-15}{(1+\omega^2)(1+4\omega^2)} - j \frac{5(1-2\omega^2)}{\omega(1+\omega^2)(1+4\omega^2)}$$

$$G(j0^+) = \infty \angle -90^\circ \quad G(j\infty) = 0 \angle -270^\circ$$

负实轴交点

$$\text{Im}[G(j\omega_g)] = 0 \rightarrow \omega_g = \frac{1}{\sqrt{2}} = 0.707$$

$$\text{Re}[G(j0.707)] = \frac{-15}{(1+0.5)(1+4 \times 0.5)} = -\frac{10}{3}$$

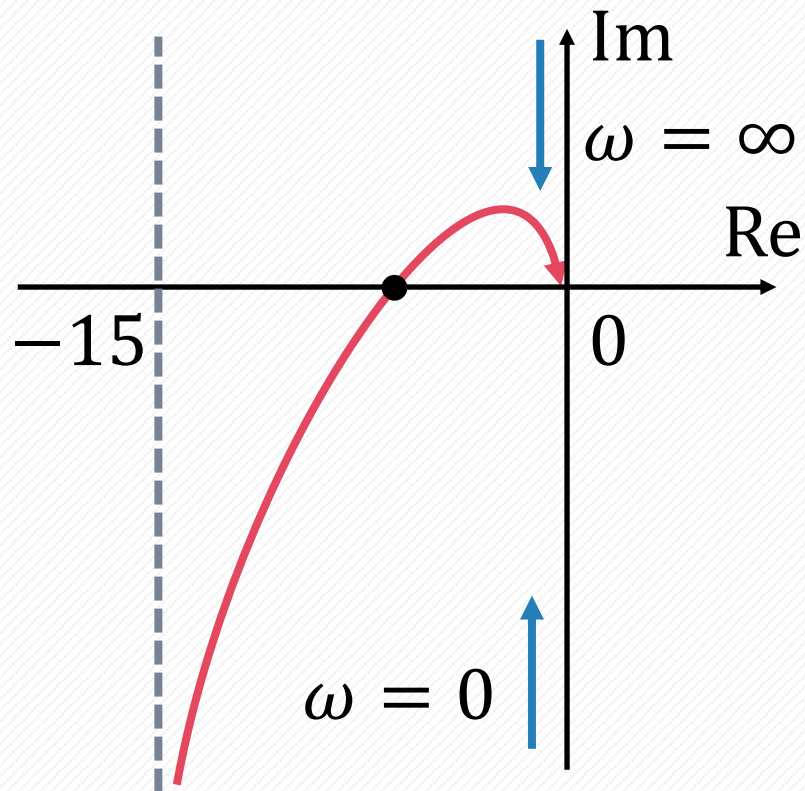
实部渐近线

$$\text{Re}[G(j0)] = -15$$

03 概略绘图

不含零点时单调收缩

$$G(s) = \frac{5}{s(s+1)(2s+1)}$$



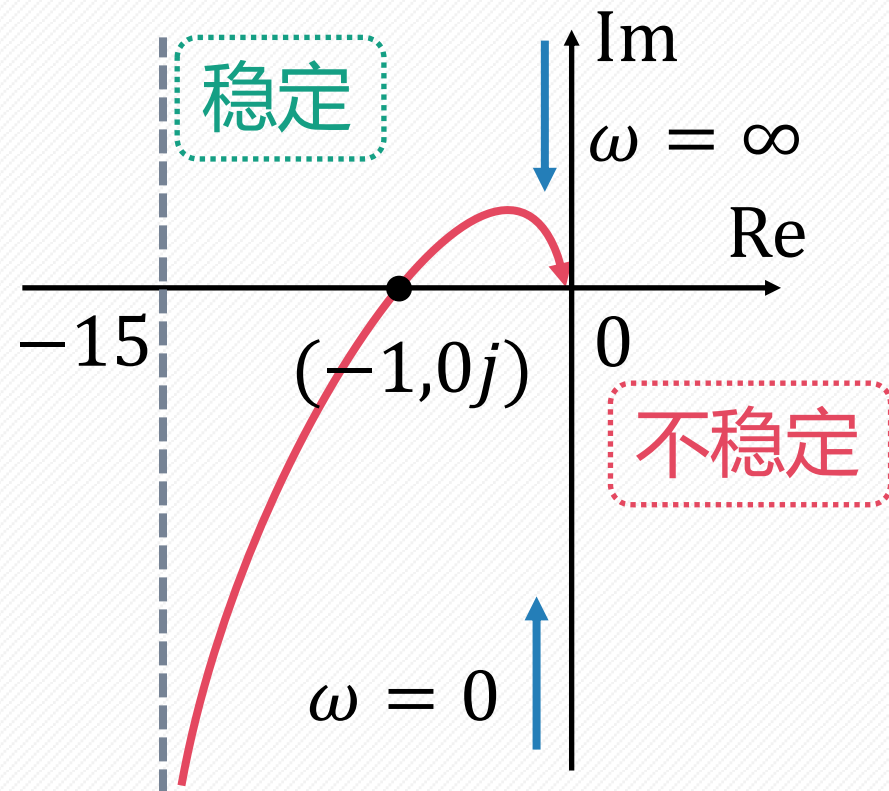
总结

频域稳定判据

开环幅相特性和 $(-1, 0j)$ 的关系

简洁下的深刻

幅角原理



总结

01 幅相特性

幅值和相角在频率 $0 \rightarrow \infty$ 变化时的极坐标轨迹

02 起点终点

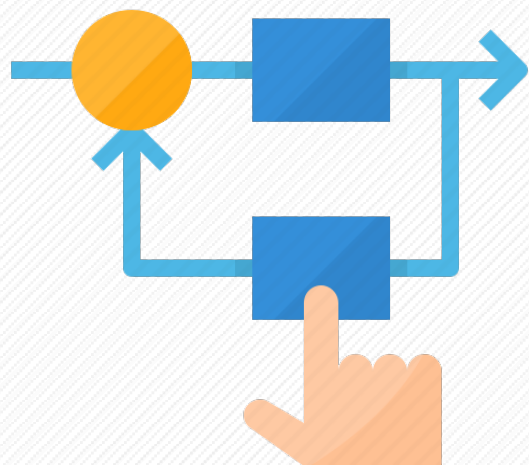
$-90^\circ \nu, -90^\circ(n - m)$

03 负实轴交点

穿越频率, $\omega_g: \angle G(j\omega_g) = -180^\circ$

04 单位圆交点

截止频率, $\omega_c: |G(j\omega_c)| = 1$



简洁下的深刻

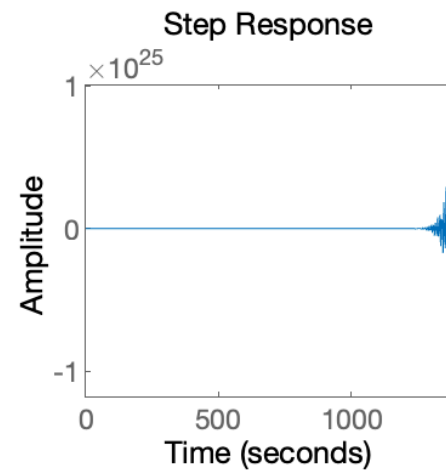
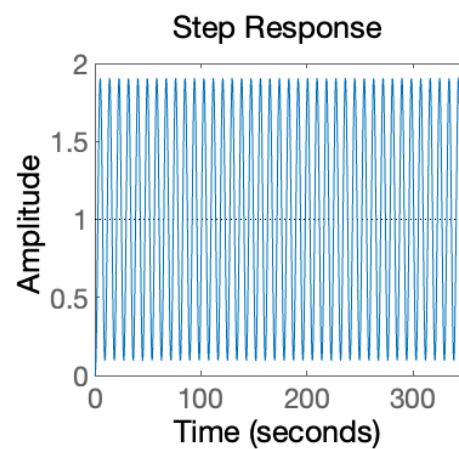
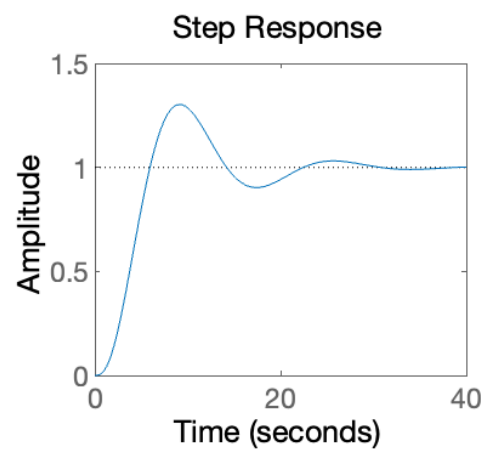
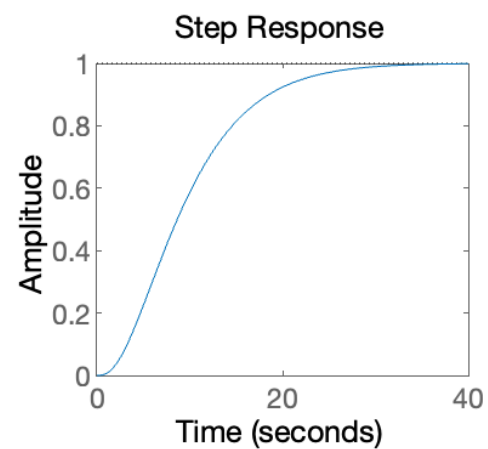
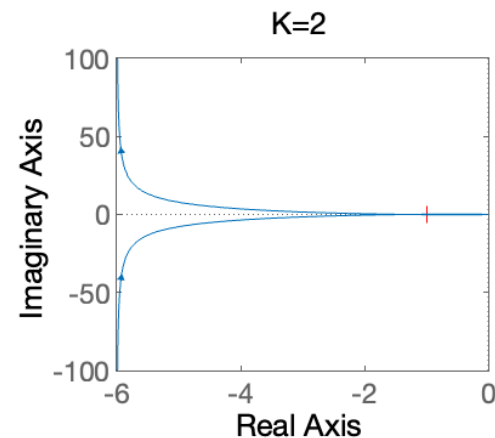
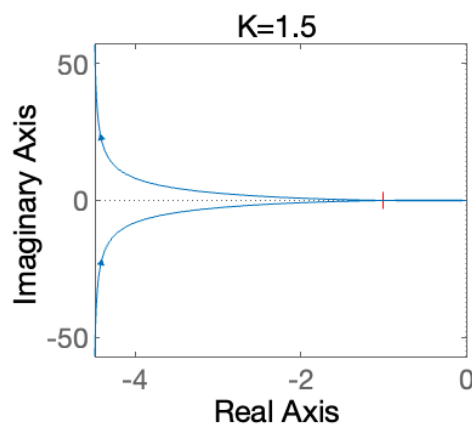
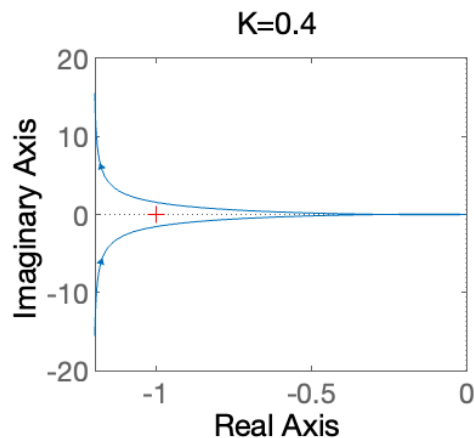
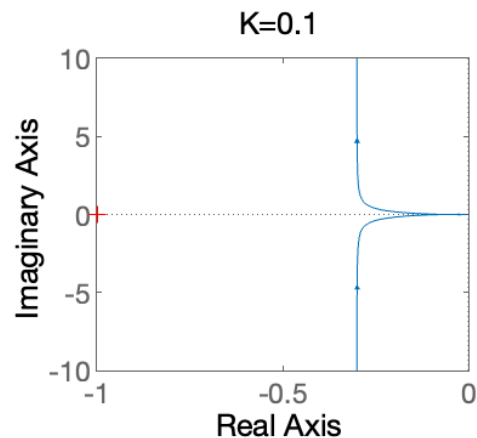
复杂现象的简洁概括

课后思考

$$G(s) = \frac{K}{s(s+1)(2s+1)}$$

?

根据下图实验数据分析开环幅相特性和闭环系统稳定性的关系



课外拓展及预习要求



DR_CAN奈奎斯特稳定性判据：

https://www.bilibili.com/video/BV1yg4y1B7JM?share_source=copy_web

- [1] Pejović, P. (2019, October). Replotting the nyquist plot-a new visualization proposal. In 2019 20th International Symposium on Power Electronics (Ee) (pp. 1-6). IEEE.
- [2] Fan, L., & Miao, Z. (2020). Admittance-based stability analysis: Bode plots, nyquist diagrams or eigenvalue analysis?. IEEE Transactions on Power Systems, 35(4), 3312-3315.

19. 幅相特性：换个角度看频域

感谢聆听！

